Where Maharashtra loses its students

Five years of school returns for every recognised school in the state — 1,08,139 schools, 2.16 crore children. Tracked grade by grade, almost none of them leave school before class 10. Then a fifth of them vanish at once, in every single year of the panel.

Source: UDISE+ Data Sharing Portal, state code 27 · up to 7 tables × 5 years · school identifiers are stable across the panel (99.3–99.7% carry-over year to year), so the same school is followed throughout

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FINDING 01 — THE EXIT POINT

Ten years of near-perfect retention, then a cliff

80.3%

of class 10 pupils in 2024-25 appear in class 11 the next year

Every other grade-to-grade step in Maharashtra holds between **99.6% and 102.6%**. The step from class 10 into class 11 holds only **80.3%** — a one-year loss of **3,49,816 children**. It is not a one-year artefact: the cliff appears in **all four** transition pairs in the panel, at 76.1%, 80.4%, 79.6% and 80.3%. Over five years it has cost roughly **15 lakh** school places.

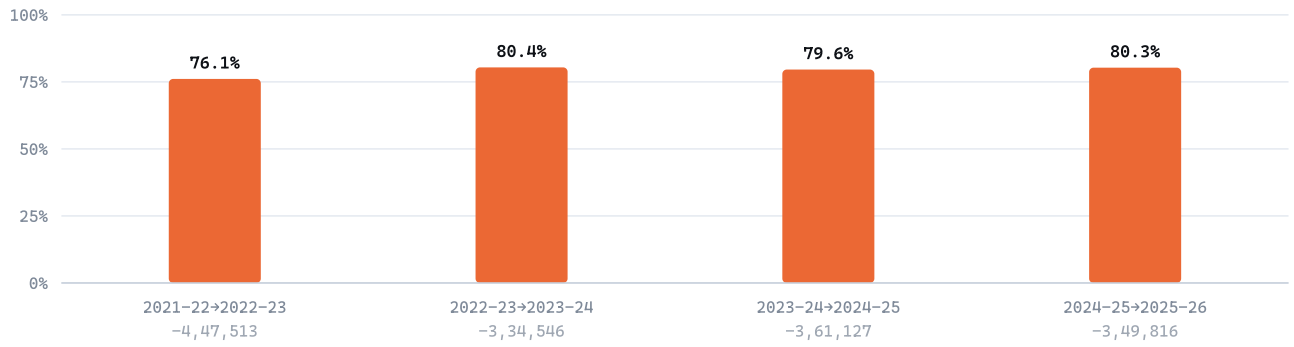
SCHOOLS	PUPILS	TEACHERS	DISTRICTS
1,08,139	2.16 crore	7,50,272	36

LOST AT CLASS 11

3,49,816 pupils

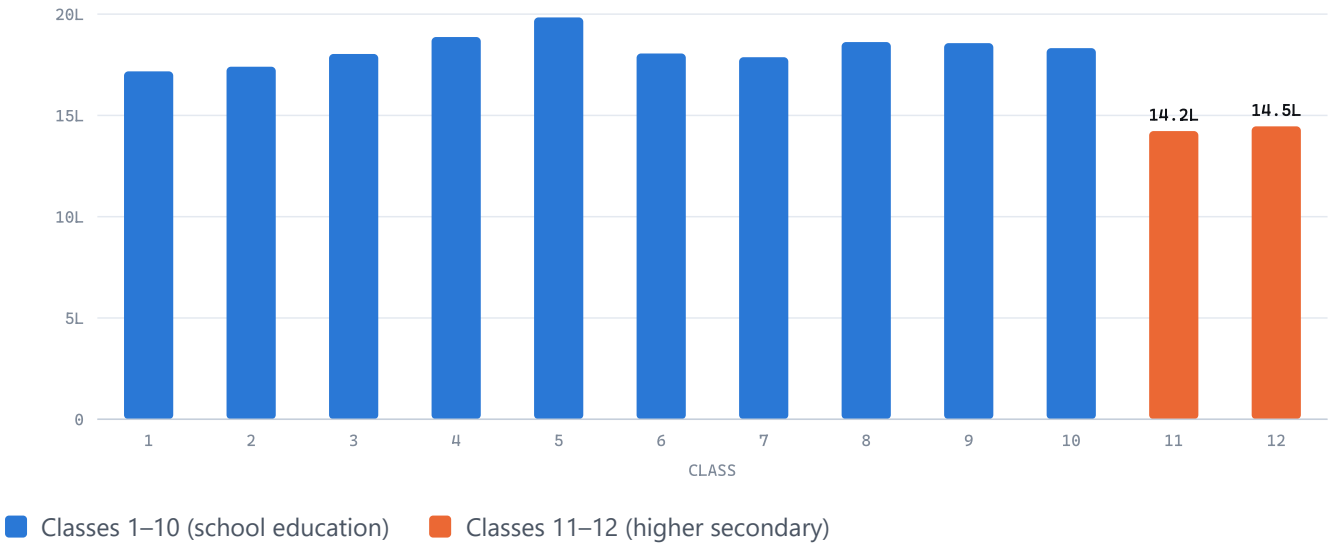
The class 10 → 11 exit, every year of the panel

Four consecutive transition pairs. The cliff is stable near 80%, which is what makes it structural rather than a shock.



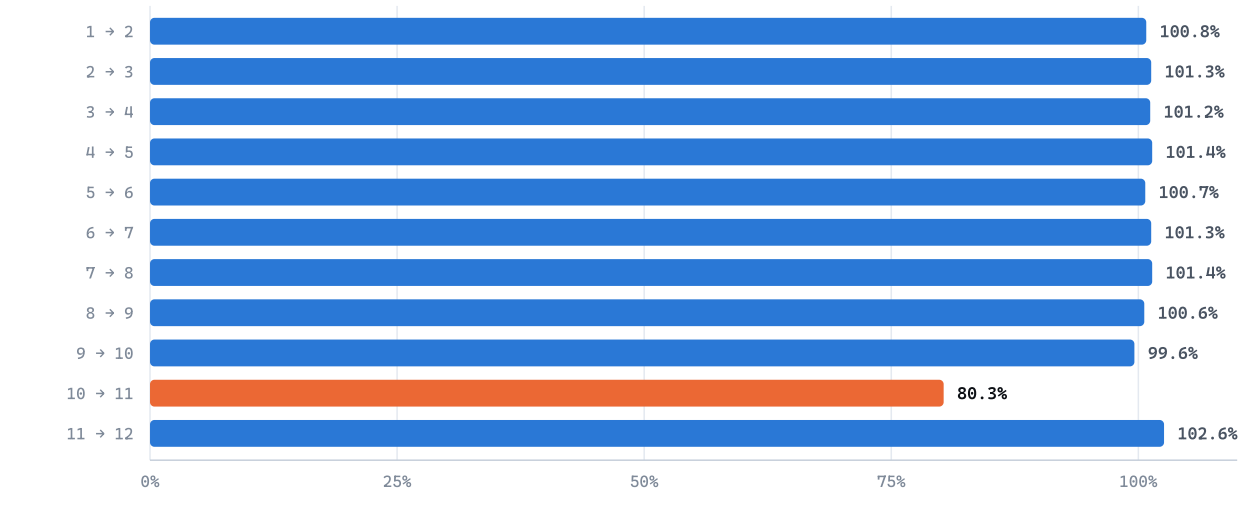
Enrolment by class, 2025-26

Classes 1 to 10 sit in a flat band near 18 lakh pupils each. Classes 11 and 12 drop to about 14 lakh — and class 11 is offered by only 10,466 schools, against 25,733 offering class 10.



Grade-to-grade retention, 2024-25 → 2025-26

Share of a grade's pupils who appear in the next grade a year later, computed on state totals. Values slightly above 100% mean the higher grade gained pupils.



FINDING 02 — WHO LEAVES

At the cliff edge, boys leave more than girls — and the gap widens every year

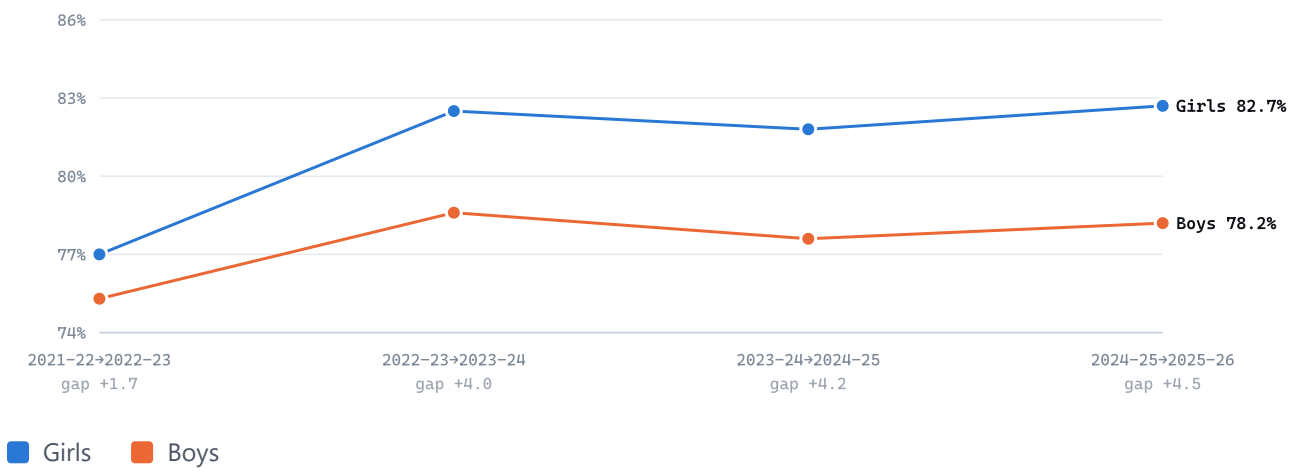
The expected pattern in Indian school data is girls dropping out faster after elementary school. Maharashtra inverts it at the decisive moment: at the class 10 → 11 exit, **girls are retained 4.5 percentage points more often than boys**.

With five years on the panel this is no longer a single reading. The gap has grown in **every consecutive pair** — 1.7, then 4.0, then 4.2, now 4.5 points — because boys' retention has fallen while girls' has held. Whatever pulls pupils out of school after class 10 in Maharashtra is pulling increasingly hard on boys.

GIRLS RETAINED, CLASS 10→11	BOYS RETAINED, CLASS 10→11	GAP, IN GIRLS' FAVOUR	GAP FOUR YEARS AGO
82.7%	**78.2%**	**+4.5 pp**	**+1.7 pp**

Class 10 → 11 retention by sex, all four transition pairs

The two lines diverge: girls' retention is broadly flat, boys' drifts down.



FINDING 03 — THE INFRASTRUCTURE GAP

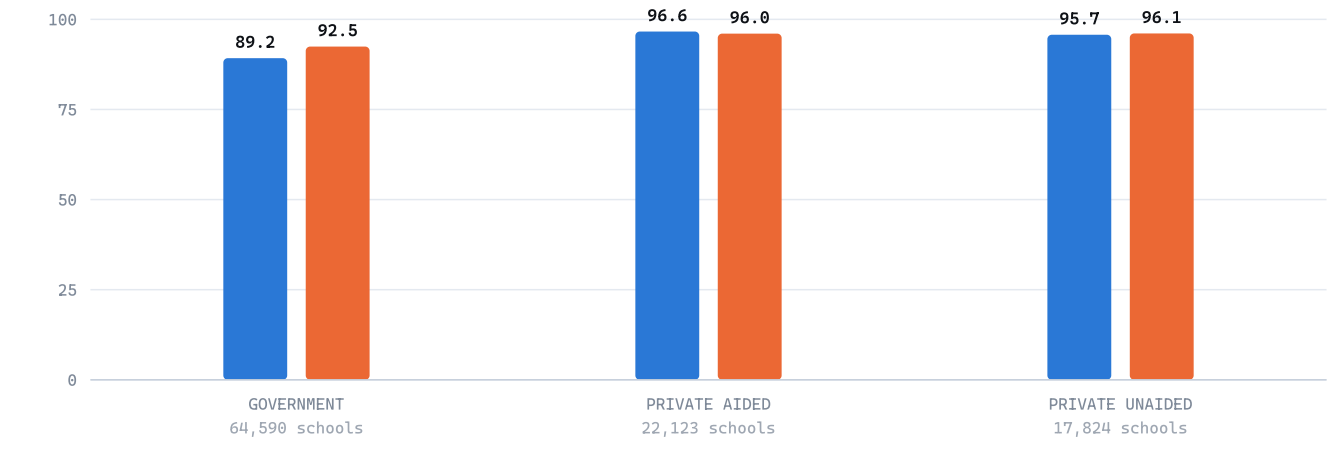
A rural government school is a different institution

Each school is scored on **11 basic facilities** it either has or lacks — functional drinking water, functional toilets for each sex it teaches, electricity, library, playground, ramps, handwashing, a computing device, internet and medical check-ups. The score is the share of applicable items met, so a boys-only school is not marked down for having no girls' toilet.

Maharashtra's average rose from **82.8 to 92.2** over the five years — real, steady improvement. But the state average hides a gap of **six and a half to seven and a half points** between the 58,247 rural government schools and the private categories around them, and rural government schools teach 37 lakh children. The gap has narrowed since 2021-22, but it has not closed.

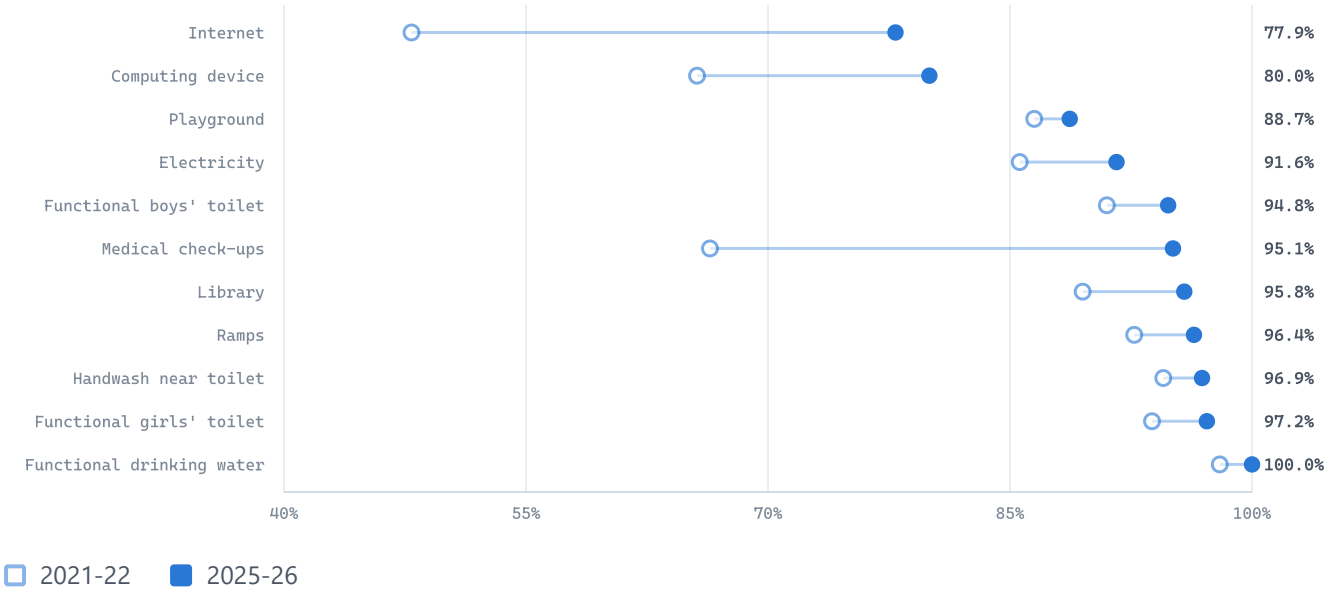
Infrastructure score by location and management, 2025-26

Mean share of applicable facilities present, 0–100.



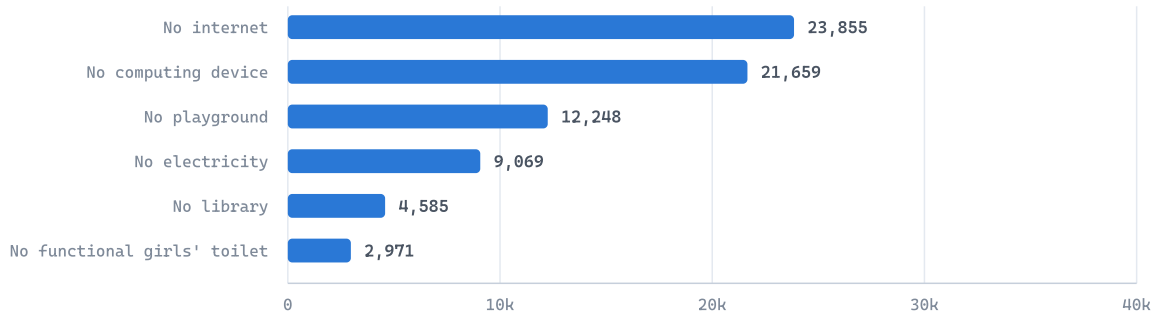
What moved over five years

Share of schools with each facility, 2021-22 → 2025-26. Connectivity moved hardest — internet up 30 points. Every indicator improved, which is itself a reason to read the last two years carefully (see the method note).



Schools still missing a basic facility, 2025-26

Counted across all 1,08,139 schools, with the pupils enrolled in them.



FINDING 04 — GEOGRAPHY

The two gaps do not line up

If poor facilities drove pupils out, the worst-equipped districts would lose the most children at class 11. They do not. Across the 35 districts where both can be measured, the correlation between infrastructure score and class 10 → 11 retention is **r = 0.26** — weak, and in the counter-intuitive direction in the clearest cases: **Jalgaon** is among the best-equipped districts in the state and *worst* on retention (62.8%), while **Gadchiroli** ranks last on infrastructure yet retains above the state average.

So the cliff is not an infrastructure problem. Five years of spending moved the facility score up nine points and left it exactly where it was. The next section tests what does explain it.

All 36 districts, 2025-26. Sorted by infrastructure score, lowest first.

DISTRICT	SCHOOLS	PUPILS	INFRA SCORE ↑	INTERNET %	POWER %	GIRLS' TOILET %	CL. 10→11 %
Gadchiroli	1,969	1,90,952	82.4	65.0%	71.5%	89.0%	76.5%
Nandurbar	2,099	3,43,182	82.5	51.1%	65.7%	97.3%	64.0%
Bid	3,712	5,38,969	85.6	71.3%	71.7%	95.4%	82.2%
Sindhudurg	1,631	95,324	86.5	33.3%	99.6%	98.5%	86.5%
Nanded	3,710	6,40,776	87.0	70.7%	80.0%	90.8%	73.8%
Hingoli	1,330	2,18,715	87.0	60.8%	82.2%	89.2%	78.5%
Parbhani	2,103	3,82,682	87.3	67.5%	83.6%	93.2%	75.1%
Palghar	3,322	7,72,935	87.9	57.8%	89.6%	97.4%	77.3%
Akola	1,844	3,34,769	88.0	69.2%	96.6%	94.5%	79.5%
Chhatrapati Sambhajinagar	4,510	9,08,536	88.5	58.8%	85.2%	99.0%	79.1%
Latur	2,664	4,85,310	88.7	62.5%	90.4%	94.8%	74.3%
Chandrapur	2,455	3,67,844	89.2	54.1%	95.6%	95.3%	81.9%
Gondiya	1,638	2,27,149	89.4	53.0%	97.2%	98.2%	90.5%
Raigarh	3,552	5,58,407	89.5	57.3%	88.9%	99.9%	78.8%
Dhule	1,981	4,44,677	89.8	78.3%	85.2%	91.3%	64.6%
Jalna	2,445	4,08,849	90.3	74.9%	88.8%	94.4%	87.3%
Bhandara	1,292	1,94,085	90.9	58.1%	99.2%	96.0%	87.6%
Amravati	2,872	4,83,910	91.1	81.5%	93.3%	96.9%	76.1%
Dharashiv	1,816	2,71,324	91.2	76.4%	92.2%	94.2%	65.2%
Buldana	2,476	5,07,124	91.4	63.2%	93.4%	97.2%	76.7%
Solapur	4,701	8,11,586	92.4	78.5%	87.3%	95.7%	72.0%
Ratnagiri	2,966	2,07,373	93.0	63.1%	98.3%	99.5%	83.6%
Yavatmal	3,326	4,84,745	93.0	86.6%	89.7%	100.0%	74.7%
Washim	1,361	2,31,921	93.4	88.0%	98.5%	99.3%	86.2%
Nashik	5,563	13,25,158	93.9	80.5%	93.7%	99.3%	75.1%
Ahilyanagar	5,265	8,41,960	95.2	83.9%	92.3%	98.0%	76.6%
Nagpur	3,966	8,73,228	95.2	98.8%	98.3%	96.7%	92.6%
Wardha	1,457	2,07,789	96.1	96.2%	100.0%	99.7%	81.2%
Mumbai II	1,759	11,22,399	96.4	100.0%	100.0%	100.0%	109.5%
Mumbai (Suburban)	2,324	7,23,715	96.6	92.9%	100.0%	99.7%	—
Pune	7,399	22,94,737	96.7	90.2%	98.2%	100.0%	87.1%
Thane	4,799	17,23,241	96.7	96.3%	97.4%	100.0%	82.1%
Jalgaon	3,387	8,35,005	96.7	97.8%	96.7%	96.3%	62.8%
Sangli	2,967	4,81,033	97.5	97.4%	96.6%	97.7%	74.3%
Kolhapur	3,675	6,45,915	97.6	99.2%	96.3%	98.9%	86.4%
Satara	3,803	4,47,345	99.2	99.8%	99.7%	100.0%	80.5%

FINDING 05 — THE MECHANISM

The cliff is a doorway problem: 55% of pupils have to change school to continue

There is a sharp test available. A pupil in a school that already runs class 11 can carry straight on. A pupil in a school that *ends* at class 10 must find a different institution

— apply, travel, pay, and be admitted. The school records tell us which of the two every pupil is in.

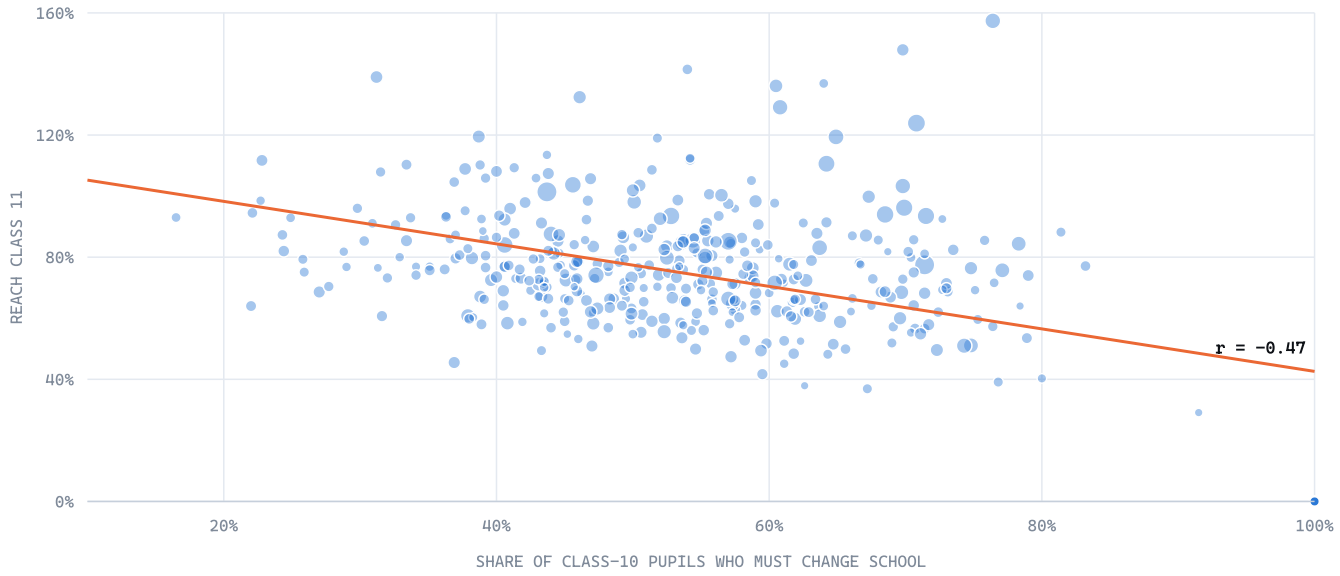
55% of all class-10 pupils — 9.7 lakh children in 17,110 schools — sit in a school with no class 11 to go on to. Meanwhile the schools that *do* offer class 11 enrol **103.7%** of the class 10 they themselves taught the year before, so they are net importers: pupils are moving towards them.

MUST CHANGE SCHOOL	SCHOOLS ENDING AT CLASS 10	CLASS-11 INTAKE AT CONTINUING SCHOOLS	PLACES IN STANDALONE JUNIOR COLLEGES
55% of class-10 pupils	**17,110**	**103.7%** of their own class 10 — net importers	**5,86,901** — arrivals, by definition

What this cannot show. A school identifier is not a pupil identifier, so a school's class-11 roll mixes its own continuing pupils with arrivals from terminal schools, and nothing in this data separates the two. That 103.7% is an *intake* ratio, not a continuation rate. The accounting identity alone bounds the two groups: between **64% and 100%** of pupils who had to move found a class-11 place, and between **56% and 100%** of those who could stay continued. It is consistent with the loss falling on either group. The one thing needing no assumption: **5,86,901** class-11 places sit in schools with no class 10 of their own, so those pupils certainly arrived from somewhere else. The block evidence below, which does not require tracking anyone, is what carries this finding.

Every block in Maharashtra: pupils who must change school vs pupils who continue

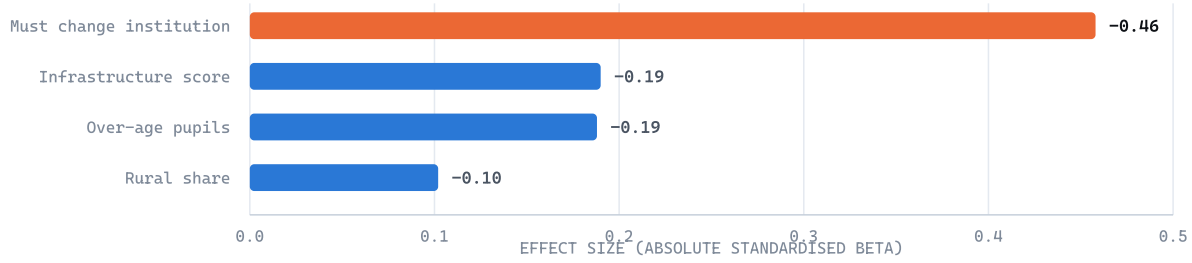
Each dot is one of 407 blocks with at least 200 class-10 pupils, sized by pupil count. The more of a block's pupils are stuck in schools that end at class 10, the fewer of them turn up in class 11 the next year.



Blocks above 100% take in pupils from neighbouring blocks — the mirror image of the ones near zero.

What actually predicts whether a block keeps its pupils

Standardised regression coefficients across 407 blocks, so the bars are directly comparable. Longer bar = stronger effect. R^2 = 0.26.



Having nowhere to go is **two and a half times** the effect of infrastructure — and infrastructure carries the wrong sign, which is the Jalgaon paradox showing up as a coefficient.

One methodological point matters here. At *district* level this relationship nearly vanishes (r = −0.08), because pupils commute across blocks inside a district and the movement cancels out. At block level it is unmistakable. The district was simply the wrong unit — which is also why the correlation in the previous section was so weak.

FINDING 06 — THE EARLY WARNING

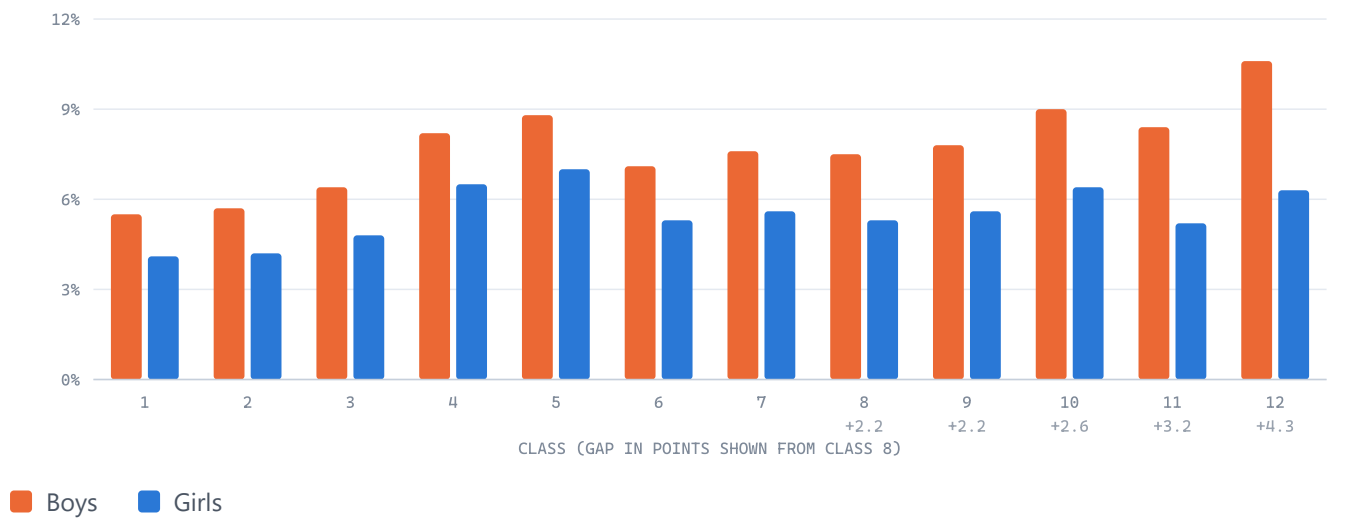
Boys arrive at the cliff already behind

The age table — 7.8 million rows, unused until now — records every pupil's age alongside their grade. Nominal age for class N is N+5, so anyone older is repeating or started late. Over-age enrolment is the standard leading indicator of dropout, and it is **rising**: 5.4% of class-10 pupils in 2022-23, **7.8%** now.

It is also strongly gendered, and the gender gap **widens exactly where the retention gap widens**. Boys are 2.2 points more likely than girls to be over-age in class 8; by class 12 the gap is 4.3 points. That is a plausible mechanism for Finding 02 — boys reach the class 11 doorway carrying more repetition, and repetition is what precedes leaving.

Over-age enrolment by grade and sex, 2025-26

Share of pupils at least one year older than the nominal age for their grade.



Across blocks, over-age share does predict lower continuation (β = −0.19) — about the same weight as infrastructure, well behind the doorway problem. **One test failed and is worth stating.** At individual school level over-age appeared to *raise* continuation, which is an artefact: schools that admit over-age pupils are receiving institutions, so their intake counts arrivals as well as their own pupils. The school-level measure cannot separate the two; only the block aggregate can.

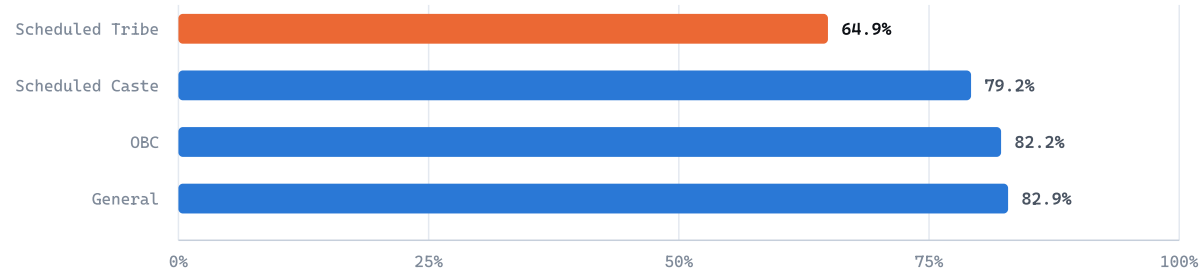
FINDING 07 — WHO FALLS OFF

Adivasi children lose the cliff by eighteen points

Splitting the class 10 → 11 step by social category gives the starkest number in this report. General-category pupils continue at **82.9%**. Scheduled Tribe pupils continue at **64.9%**. That is an **18-point gap**, and it is not a fluke of one year — ST retention has been the lowest of the four groups in every transition pair in the panel.

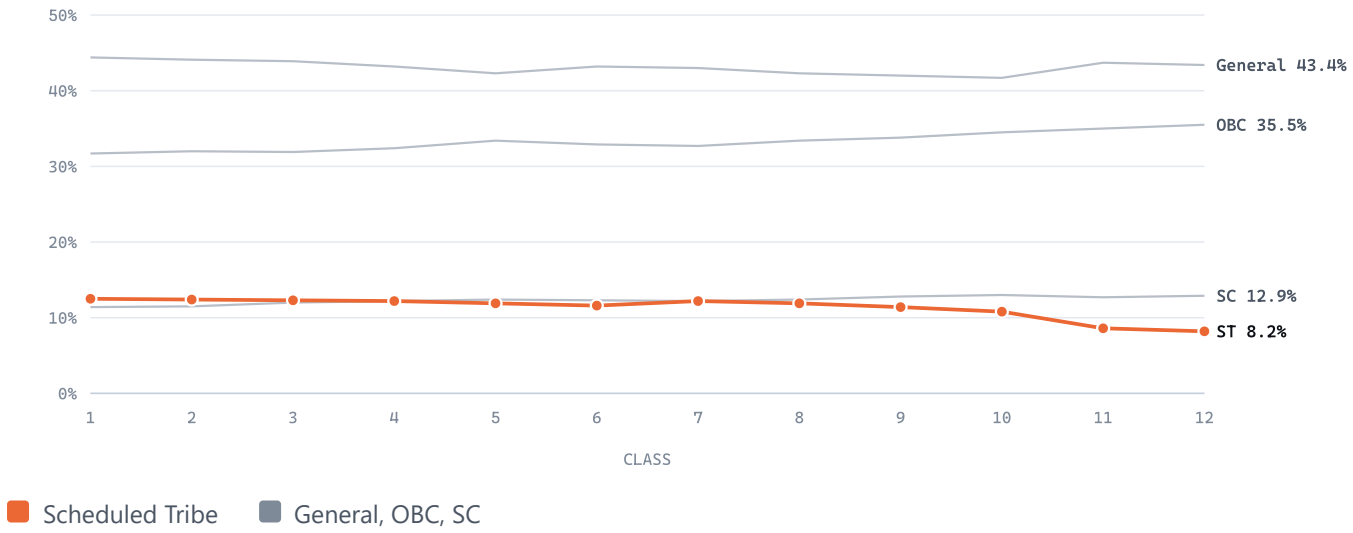
It shows up as compositional bleed. ST children are **12.5%** of class 1 and **8.2%** of class 12, and more than half of that entire loss happens in the single step from class 10 to 11.

Class 10 → 11 retention by social category, 2024-25 → 2025-26



Share of each grade by social category, 2025-26

Every group holds roughly steady through school. ST is the exception, and the drop is concentrated at the class 11 doorway.

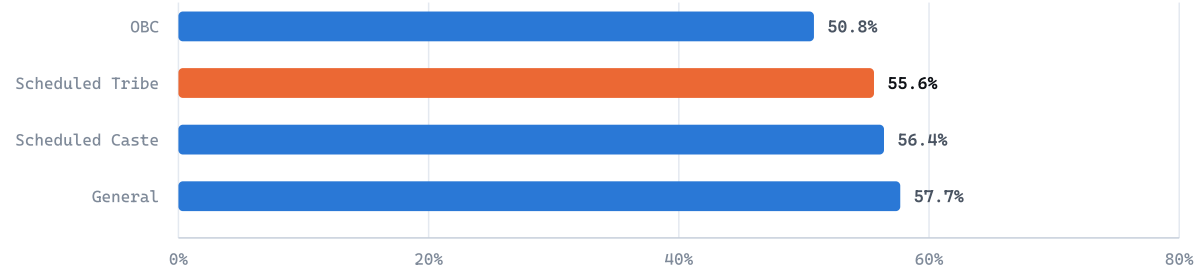


But the doorway does not explain it

The obvious hypothesis is that Finding 05 accounts for this — that ST pupils are stuck in schools that end at class 10. **They are not.** The share sitting in a terminal school is 55.6% for ST against 57.7% for General: essentially identical, and if anything slightly *lower* for ST. The doorway burden is spread evenly across every category.

Share of class-10 pupils in a school that ends at class 10

Near-identical across categories — so this is not the cause of the caste gap.



So there are **two separate mechanisms**. The doorway explains *where* the cliff is — which blocks lose pupils. Caste explains *who* goes over it. A useful clue for the second: ST class-10 pupils are **78% rural** against 35% for General pupils, so even where a junior college exists somewhere in the block, it is a different journey.

The two compound where you would fear. In **Jalgaon** — already the worst district in the state at 62.8% — ST retention is **42.5%**. In Dhule it is 46.3%. Nandurbar, where ST pupils are 72.7% of the class-10 cohort, retains 56.3%.

One last comparison puts the earlier gender finding in proportion. Girls out-retain boys inside every category — the gap is widest for SC pupils, at 6.8 points. But **ST girls continue at 67.0%, still fourteen points below General boys at 81.2%.** On this transition, caste outweighs sex by a wide margin.

FINDING 08 — STAFFING

The teacher shortage is not where you would look for it

Across elementary schools the state's pupil-teacher ratio is comfortably inside the RTE norm of 30:1 — but the aggregate conceals two opposite failures. Urban private-aided schools run the **heaviest classrooms in the state** (35.9:1). Rural government schools run the lightest ratio (19.3:1) while containing **8,152 single-teacher schools** — a school where one adult teaches every grade. That count has risen in every year of the panel, from 5,906 in 2021-22 to 9,269 statewide now, even as total teachers rose.

Elementary-stage schools only, 2025-26.

SEGMENT	SCHOOLS	PTR ↓	OVER 30:1	SINGLE-TEACHER
Urban · Private Aided	5,445	35.9	2,671	181
Urban · Private Unaided	5,545	31.8	2,308	93
Urban · Government	5,592	30.3	1,832	413
Rural · Private Aided	4,481	28.5	1,622	147
Rural · Private Unaided	3,323	21.6	651	69
Rural · Government	57,184	19.3	4,401	8,152

FINDING 09 — NEW THIS YEAR

The first school-safety return shows the sharpest divide in the data

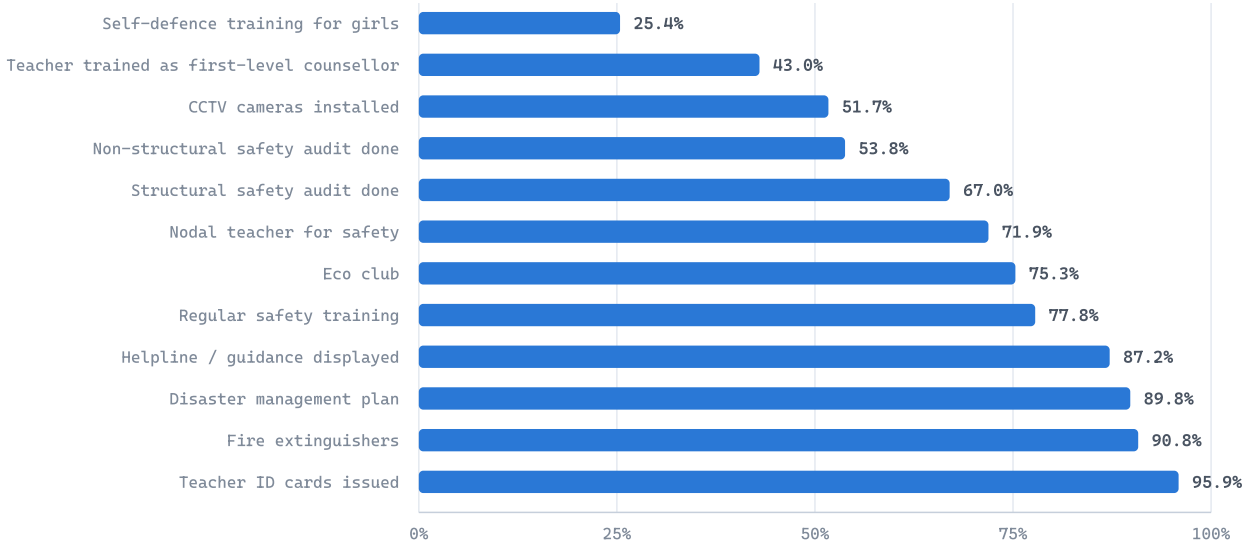
2025-26 adds a table that has never been collected before: school safety. Fire extinguishers, disaster plans and teacher ID cards are close to universal. The measures that need money or trained staff are not.

CCTV splits the state almost perfectly along the line this report keeps finding. 92.6% of urban private unaided schools have cameras; 27.7% of rural government

schools do — a **65-point gap**, the widest of any indicator in five years of data. Only **25.4%** of schools report self-defence training for girls, and only 43% have a teacher trained as a first-level counsellor.

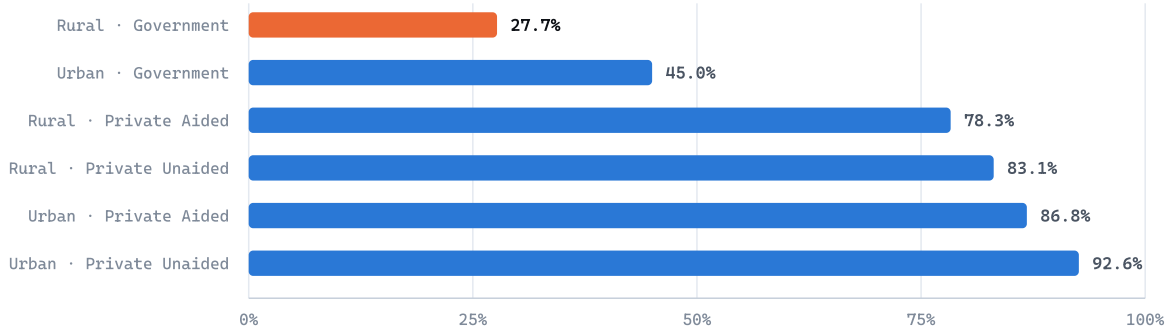
School safety measures, 2025-26

Share answering yes, out of schools answering yes or no. Self-defence training is the one measure with a large “not applicable” group (49,526 schools), so its base is smaller.



CCTV coverage by location and management, 2025-26

The same rural-government gap, at its widest.



METHOD

What the data does and does not support

Six tables per year (seven in 2025-26) — profile, facilities, teachers, safety and three enrolment files — joined on the school identifier into one panel of 5,42,682 school-year rows across three different UDISE+ release schemas, plus 7.8 million rows of age-by-grade enrolment. The age table reconciles exactly with the headcount table for every grade, which is the check that it is joined correctly. Regressions are ordinary least squares on 407 blocks with at least 200 class-10 pupils. Enrolment totals are built from the four social categories (General, SC, ST, OBC), which partition every pupil exactly once. The rebuilt totals reproduce Maharashtra's published UDISE+

figures (2.16 crore pupils, PTR 28.8 in 2025-26), which is the check that the joins are right.

SCHEMA **Three release formats across five years**

2021-22 misspells the key column as `psuedocode`, keys enrolment by a text label instead of numeric codes, stores one drinking-water flag where later years store six, and omits the teacher caste breakdown. 2022-23 to 2024-25 share one schema. 2025-26 changes again: facilities gain 12 columns, profile gains 11, and enrolment splits transgender pupils into their own `*_t` column — earlier years folded them into `*_b`, so they are folded back here to keep the series continuous (statewide that is 1–2 pupils).

RECLASSIFICATION **Management code 7 is not what it looks like**

In 2025-26 management code 7 jumps from 79 schools to 3,402. Tracking those schools back one year shows 2,359 were Private Aided and 962 Private Unaided, so it is a private-sector residual category, not a government one. It is kept in its own group here; folding it into "Government" on the strength of a standard code list would have moved 3,300 private schools across this report's central comparison.

COMPARABILITY **2021-22 totals, and two suspiciously perfect trends**

Total enrolment falls from 2.26 crore to 2.12 crore between the first two years, pushing every 2021-22 transition down to roughly 94%. That is a reporting-basis change, not 14 lakh children leaving school in one year, so headline retention uses the latest pair only. Two later series need the same scepticism: the "not applicable" water code was retired in 2024-25, so functional drinking water reaching 100% is a coding change rather than an overnight fix; and schools recording zero teachers fall from 943 to 9, which is data validation, not hiring. Single-teacher schools — which rose over the same period — are the more trustworthy staffing measure.

TRACKING **Schools can be followed; pupils cannot**

`pseudocode` is a school identifier, not a pupil one, and there is no pupil-level file. Every "retention" figure here is therefore a ratio of two headcounts a year apart, never a followed cohort. This matters most in Finding 05: a school's class-11 roll mixes its own continuing pupils with arrivals, so no school-level ratio can be read as a continuation rate. Only aggregate bounds and the block model survive that limitation, and only those are relied on.

CAUSATION **What the block model does and does not show**

Block retention counts pupils who reach class 11 *in that block*, so a pupil who commutes to the next block registers as a local loss. The block model therefore establishes that local supply drives local continuation; it does not by itself prove that the 3.5 lakh statewide shortfall is dropout. The statewide figure is the one that nets out commuting, and it is the one quoted as the loss. Over-age enrolment for 2021-22 reads 26.4% against 5–8% in every later year, so that year is excluded from the age series along with everything else about it.

COHORTS

Aggregate, never school-by-school

Maharashtra's schools mostly end at class 4 or 7, so pupils change school at those boundaries. Tracking a cohort within one school would count a normal transfer as a dropout. All retention is computed on state or district totals, where transfers net out.

DISTRICTS

Three renames, one gap

Aurangabad became Chhatrapati Sambhajinagar, Osmanabad became Dharashiv, and Ahmadnagar became Ahilyanagar in 2025-26; all three are merged here, each confirmed by matching school counts. Mumbai (Suburban) reports no class 11–12 at all — its junior colleges sit under Mumbai II, which is why Mumbai II shows a retention above 100% and Mumbai (Suburban) shows none.

EXCLUDED

Columns that could not be made comparable

`furniture_availability` is a count of furniture sets in 2021-22 (0–18,777) and a 1/2/3 code from 2022-23 onward, so no single test works across the panel. `boundary_wall` and `building_status` are type codes whose labels appear in no supplied schema PDF. All three are left out of the infrastructure index rather than guessed at.

LIMITS

What is not in this data

School identifiers are pseudonymised and there are no coordinates, so no school can be named or mapped below district level. There are no learning outcomes, no attendance and no finance beyond a grant receipt flag — so “retention” here means appearing in the next grade's count, not passing anything.

Built from the raw UDISE+ zips with DuckDB. Pipeline: `01_extract` → `04_build_panel` → `06_analyse`. Every figure on this page is reproducible from `data/udise.duckdb`.